Separable quintic: affine and projective point domains

Formal proof packet

28 October 2026

This packet records a checked theorem with a visible separability hypothesis. It contains no experiment, target result, or cost ratio. The Lean source, compiler log, editable SVG, and CSV gate table accompany this PDF.

Checked statement and argument

Let K be any field and $f(X) = X^5 + a_3X^3 + a_2X^2 + a_1X + a_0$. If f is separable, the pinned Lean proof extracts $t, u \in K[X]$ with $f = tu$, t prime, and $t \nmid u$. The quintic has degree five and is not a unit, so it has an irreducible factor. Separability supplies squarefreeness and excludes that factor from the cofactor. The existing checked Eisenstein lemma then makes $Y^2 - f(X)$ irreducible over $K[X][Y]$.

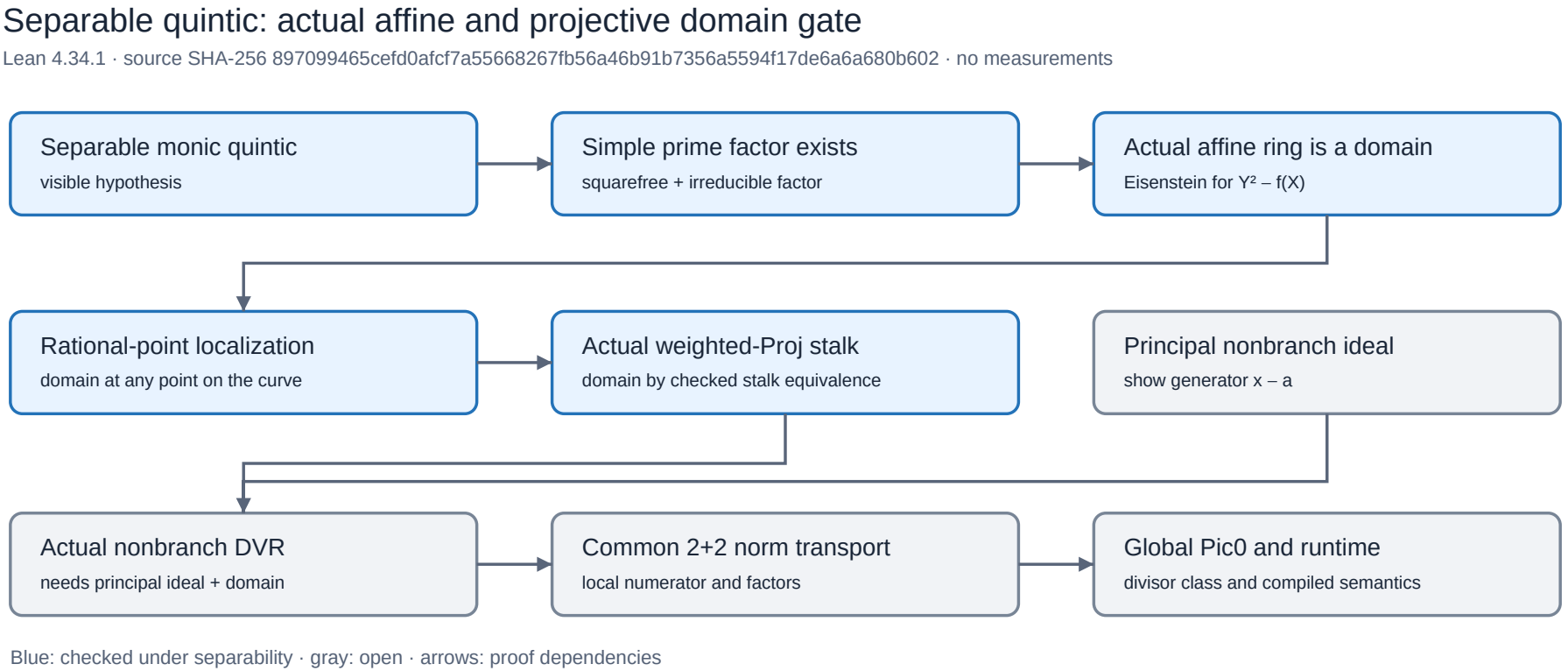
The new theorems establish that $K[X,Y]/(Y^2 - f(X))$ is a domain, its localization at any rational point satisfying $y^2 = f(a)$ is a domain, and the corresponding actual weighted-Proj point stalk is a domain. The one-variable equivalence `MvPolynomial.uniqueAlgEquiv` identifies the extracted factor with the actual quintic representation. No characteristic assumption is needed for this domain statement. The earlier local norm-order theorem still requires its stated hypotheses, including an actual DVR.

Verification boundary

Direct compilation used pinned Lean 4.34.1 and Mathlib revision
`d13f23b723b8a846827a245b89c10fc7d3f11612`. The command `TMPDIR=/dev/shm lake env lean Common22SeparableAffineDomain.lean` exited zero. All seven printed axiom audits list only `propext`, `Classical.choice`, and `Quot.sound`; no `sorryAx`. Two nonfatal style-linter messages recommend `let` in place of `letI`. Lean source SHA-256:
`897099465cefd0afcf7a55668267fb56a46b91b7356a5594f17de6a6a680b602`.

The exact next local geometry gate is principality of the evaluation maximal ideal in the nonbranch stalk; the checked domain alone does not establish a DVR or order formula. Formula-specific norm transport, global principal-divisor/Pic^0 transport, and compiled Rust semantics are open. An odd-degree proof without separability is also open in this formal line. The CSV preserves these statuses for independent review.

Proof dependencies and open gates



Blue nodes are checked under separability. Gray nodes are open. Arrows show proof dependencies, not measured performance.